

Targeted SAT Math — Set 02

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Homework after Test 11. Work on paper. Calculator / Desmos is fine.

30 questions. These are only the leaks from Targeted Set 01 and Test 11. Items are instructor-written SAT-style questions, **not** official College Board stems, and **not** taken from Practice Tests 1–11.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

1. Notes for this set only

Successive percent

- Each change **multiplies**. Do not add the percents.
- Up $p\%$ means $\times \left(1 + \frac{p}{100}\right)$. Down $p\%$ means $\times \left(1 - \frac{p}{100}\right)$.
- After you multiply, you already have the new amount. **Do not add the starting amount back on**. The grown number includes the start.
- Two identical ups: square the factor. +20% then +20% is $\times 1.44$, a **44%** increase, not +40%.

Tangent to a circle — 0 vs undefined

- The radius to the contact point P is **perpendicular** to the tangent at P .
- A line **perpendicular to the x-axis** is vertical. Its slope is **undefined**.
- A line **parallel to the x-axis** is horizontal. Its slope is **0**.
- Same x on center and $P \Rightarrow$ radius is vertical $\Rightarrow$ tangent is horizontal $\Rightarrow$ slope **0**.
- Same y on center and $P \Rightarrow$ radius is horizontal $\Rightarrow$ tangent is vertical $\Rightarrow$ slope **undefined**.
- SAT asks for the **tangent**, not the radius. Circle that word before you bubble.

Right-triangle trig

- Name the angle first. Opposite / adjacent / hypotenuse are relative to **that** angle.
- $\sin \theta = \frac{\text{opp}}{\text{hyp}}$, $\cos \theta = \frac{\text{adj}}{\text{hyp}}$, $\tan \theta = \frac{\text{opp}}{\text{adj}}$. Isolate the side they asked for.

- If the sides are ugly (165 and 220, 156 and 208), SAT wants the **ratio**. Those numbers cancel. Do **not** Pythagorean a giant hypotenuse first.

Messy estimate

- The graph or intercept is allowed to be a little dirty. Compute, then take **closest**.
- Write what the intercept means (all of one type, none of the other) before you divide.
- The other intercept is the usual trap.

Linear systems — how many solutions

- Same line (one equation is a scale of the other) $\Rightarrow$ infinitely many.
- Same slope, different intercept $\Rightarrow$ none.
- Different slopes $\Rightarrow$ exactly one.
- “At least one” means infinitely many **or** exactly one.

Area units

- Length converts with k . Area converts with k^2 .
- Square the linear factor **before** you multiply by the given area.

Two lines, four angles

- Verticals are equal. Adjacent angles sum to 180° .
- There are only three possible pairwise sums. “Could NOT” means the choice that is not one of those three.

2. Questions

1. (SPR) A population starts at 120 and grows by 25% each year. What is the population after 2 years?

Answer: __

2. Which value is equivalent to two successive 20% increases?

A) a 40% increase

B) $\times 1.40$

C) $\times 1.44$

D) $\times 1.20$

3. (SPR) A population starts at 160 and grows by 25% each year. What is the population after 2 years?

Answer: __

4. A circle has center $(5, 2)$. A line is tangent to the circle at $(5, 9)$. What is the slope of the tangent line?

A) 0

B) 7

C) undefined

D) 1

5. (SPR) A circle has center $(1, -3)$. A line is tangent to the circle at $(1, 4)$. What is the slope of the tangent line?

Answer: __

6. A circle has center $(4, 6)$. A line is tangent to the circle at $(11, 6)$. What is the slope of the tangent line?

A) 0

B) 7

C) undefined

D) 1

7. A circle has center $(-2, 3)$. A line is tangent to the circle at $(5, 3)$. What is the slope of the tangent line?

A) 0

B) undefined

C) $\frac{7}{5}$

D) $-\frac{5}{7}$

8. In right triangle ABC , the right angle is at C , and $AC = 12$. Which expression represents the length of BC ?

A) $12 \sin A$

B) $12 \tan A$

C) $\frac{12}{\tan A}$

D) $12 \cos A$

9. (SPR) In right triangle ABC , the right angle is at C , $AC = 15$, and $BC = 8$. What is $\tan A$?

Answer: __

10. In right triangle PQR , the right angle is at R , $PR = 165$, and $QR = 220$. What is $\tan P$?

- A) $\frac{4}{3}$
- B) $\frac{3}{4}$
- C) $\frac{4}{5}$
- D) $\frac{3}{5}$

11. A grove contains only type-X trees and type-Y trees. The estimated total mass of the trees is 14,820 kilograms. A graph models the possible combinations of the number of type-X trees, x , and type-Y trees, y . Based on the graph, the x -intercept is approximately $(36, 0)$. Which of the following is closest to the estimated mass, in kilograms, of each type-X tree?

- A) 355
- B) 412
- C) 530
- D) 14,820

12. The equation $x + 3y = 12$ is one equation in a system of two linear equations. If the system has at least one solution, which of the following could be the other equation?

I. $2x + 6y = 24$

II. $2x + 6y = 20$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

13. (SPR) An area of 25 square yards is equivalent to k square meters. To the nearest tenth, what is the value of k ?

1 yard = 0.9144 meters

Answer: __

14. Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(70 - 3x)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

- A) $(140 - 6x)^\circ$
- B) $(220 + 6x)^\circ$

C) 180°

D) $(140 + 6x)^\circ$

15. The graph of $y = f(x) + 6$ is the line $y = 2x + 1$. Which equation defines f ?

A) $f(x) = 2x + 7$

B) $f(x) = 2x - 5$

C) $f(x) = 2x + 1$

D) $f(x) = 2x - 1$

16. A \$90 price increases by 20%, then the new price decreases by 20%. What is the final price?

A) \$90

B) \$86.40

C) \$88

D) \$72

17. A radius of a circle is parallel to the x-axis. What is the slope of the tangent line at that contact point?

A) 0

B) undefined

C) 1

D) -1

18. In triangle XYZ , the measure of angle X is 90° . Point W lies on segment YZ , and segment XW is perpendicular to segment YZ . The length of segment WY is 208, and the length of segment XW is 156. What is the value of $\tan Z$?

A) $\frac{3}{4}$

B) $\frac{3}{4}$

C) $\frac{3}{5}$

D) $\frac{3}{5}$

19. A workshop contains only type-A machines and type-B machines. The estimated total mass of the machines is 6,375 kilograms. A graph models the possible combinations of the number of type-A machines, x , and type-B machines, y . Based on the graph, the y -intercept is approximately $(0, 26)$. Which of the following is closest to the estimated mass, in kilograms, of each type-B machine?

A) 213

B) 245

C) 319

D) 160

20. A system of two linear equations includes $4x - 2y = 10$ and $2x - y = 5$. How many solutions does the system have?

A) none

B) exactly one

C) infinitely many

D) exactly two

21. (SPR) An area of 8 square miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k ?

1 mile = 1.609 kilometers

Answer: __

22. (SPR) A club has 50 members. Membership increases 20% in year 1 and increases 20% again in year 2. How many members are there after 2 years?

Answer: __

23. A circle has center $(0, 8)$. A line is tangent to the circle at $(6, 8)$. What is the slope of the tangent line?

A) 0

B) undefined

C) $\frac{4}{3}$

D) $-\frac{3}{4}$

24. In right triangle ABC , the right angle is at C , and hypotenuse $AB = 18$. Which expression represents the length of AC ?

A) $18 \cos B$

B) $18 \sin B$

C) $\frac{18}{\sin B}$

D) $18 \tan B$

25. Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(50 + x)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

A) $(100 + 2x)^\circ$

B) 180°

- C) $(260 - 2x)^\circ$
D) $(100 - 2x)^\circ$

26. The equation $5x + y = 8$ is one equation in a system of two linear equations. If the system has at least one solution, which of the following could be the other equation?

I. $10x + 2y = 16$

II. $x - y = 1$

- A) I only
B) II only
C) I and II
D) Neither I nor II

27. The function f is defined by $f(x) = -2x^2 + 12x - 7$. Which equivalent form displays the maximum value of the function as a constant?

- A) $f(x) = -2x(x - 6) - 7$
B) $f(x) = -2(x - 3)^2 + 11$
C) $f(x) = -2x(x - 6) + 7$
D) $f(x) = 2(x - 3)^2 - 11$

28. (SPR) A quantity is increased by 50%, then that result is increased by 50% again. The final value is 180. What was the original quantity?

Answer: __

29. A circle has center $(7, 1)$. A line is tangent to the circle at $(2, 1)$. What is the slope of the tangent line?

- A) 0
B) undefined
C) 5
D) $\frac{1}{5}$

30. (SPR) In right triangle DEF , the right angle is at F , $DF = 140$, and $EF = 105$. What is $\tan D$?

Answer: __